

Artificial Intelligence in Healthcare Waste Management: A Narrative Review of Current Applications and Future Prospects

Nurhasmadiar Nandini¹, Rani Tiya Budiyanti¹, and Sutopo Patria Jati¹

¹Health Policy and Administration, Faculty of Public Health, Universitas Diponegoro, Semarang, Indonesia

Abstract. Effective waste management in healthcare is crucial to ensure patient safety, public health, and environmental sustainability. The escalating volume of medical waste requires innovative, efficient, and sustainable management strategies. Artificial Intelligence (AI) has a significant potential to enhance waste management systems within the healthcare sector. This article aimed to explore the application of AI in healthcare waste management while assessing the associated challenges and opportunities. This study was a literature review using a narrative review approach. AI can be utilized in various aspects of waste management, including automated waste sorting and classification, waste volume prediction, waste collection optimization, and regulatory compliance monitoring. However, its implementation faces challenges, such as limited system integration, insufficient high-quality data, and ethical and regulatory concerns. Collaboration between academia, healthcare practitioners, and policymakers is essential for the development of AI applications for safe and sustainable healthcare waste management.

1 Introduction

The management of healthcare waste (HCW) remains a significant challenge in the healthcare sector. HCW encompasses all forms of waste produced by health services, including both hazardous and non-hazardous waste as well as infectious and non-infectious waste. Approximately 85% of this waste is non-hazardous, whereas the remaining 15% is hazardous, infectious, and may contain radioactive or toxic substances [1]. This hazardous waste can harbor harmful microorganisms that are capable of spreading and causing serious illnesses.

With the increasing human population and expansion of health facilities, the number of healthcare workers is also rising globally [2]. According to a 2012 WHO report on 22 developing nations, between 18% to 64% of healthcare facilities are inadequately handling HCW [3]. Healthcare waste (HCW) poses potential hazards and acts as a pollutant to

¹ Corresponding author: ranitiyas@lecturer.undip.ac.id

individuals in proximity to or in contact with medical waste, including doctors, nurses, healthcare support staff, patients, visitors to healthcare facilities, and support service workers such as laundry workers, waste management and transportation staff, and waste disposal facility employees [4]. Generally, more than two million healthcare workers are exposed to pathogens because of their daily work routines [5]. Therefore, it is imperative to formulate effective waste management principles to mitigate the adverse effects of medical waste.

Artificial intelligence holds significant potential for application in waste management. Machine learning has been employed to predict waste-generation patterns [6]. Furthermore, AI-driven waste separation technologies have been extensively developed to autonomously categorize various types of waste, thereby enhancing the efficiency of the waste collection and recycling processes [7]. Consequently, this study aimed to investigate the application of AI in waste management in healthcare services. Through this narrative review the application of AI in healthcare waste management and future directions will be explored.

2 methods

Research Type

The research methodology employed was a literature review utilizing a narrative approach, focusing on the application of artificial intelligence (AI) in waste management within healthcare facilities.

Data Source

The Scopus database was utilized, with specific keywords, including 'artificial intelligence' OR 'machine learning' AND 'waste management' AND 'healthcare.' All of the articles including books, articles, reviews, and editorial related to the topic are collected.

Inclusion and exclusion criteria

The inclusion criteria specified English-language studies published between 2015 and 2025, while excluding articles that were not related to this study

Data analyze

The selection data will be analyzed and presented in narrative form

3 Results and discussion

According to the search results from the Scopus database, 121 articles were identified as relevant to the topic under investigation, with 118 articles published within the specified timeframe from 2015 to 2025. All articles were written in English. Additionally, 106 articles did not pertain to the application of artificial intelligence in waste management within health services. Consequently, 12 articles were identified for further discussion (Figure 1).

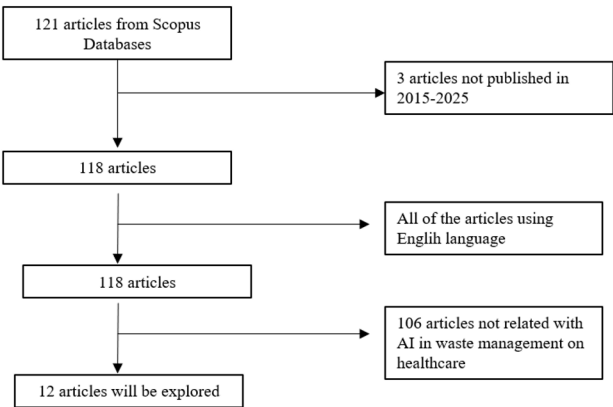


Fig 1. Search results of articles from Scopus Databases

All articles related to leveraging AI in waste management in healthcare are listed in Table 1).

Table 1. Articles related to the leveraging of AI in waste management in healthcare

No	Title	Year	Author
1	Bibliometric analysis of the applicability of artificial intelligence in the integrated management of medical waste	2024	Carrazco, Diego Cajamarca,Tobar-Ruiz, María Gabriela,López, Diego Marcelo Almeida,Hermida, Carlos Eduardo Cevallos,Arellano, Verónica Magdalena Llangari,Tobar, Mateo Augusto Zavala;Godoy, María Magdalena Paredes
2	Artificial intelligence for waste management in smart cities: a review	2023	Fang, Bingbing, Yu, Jiacheng,Chen, Zhonghao,Osman, Ahmed I, Farghali, Mohamed, Ihara, Ikko, Hamza, Essam H, Rooney, David W, Yap, Pow-Seng
3	Towards blockchain-based hospital waste management systems; applications and future trends	2022	Bamakan, Seyed Mojtaba Hosseini, Malekinejad, Pooria;Ziaeiian, Mehran

4	Artificial Intelligence-Driven Technologies for Environmental Sustainability in the Healthcare Industry	2024	Dewasiri, Narayanage Jayantha; Rathnasiri, Mananage Shanika Hansini; Karunarathna, Karunarathnage Sajith Senaka Nuwansiri
5	A review on digital transformation in healthcare waste management: Applications, research trends and implications	2025	Sharma, Vaibhav; Jamwal, Anbesh; Agrawal, Rajeev; Pratap, Saurabh
6	A critical analysis of the role of artificial intelligence and machine learning in enhancing nuclear waste management	2024	Chenniappan, Thiagarajan ; Devarajan, Yuvarajan
7	Exploring the potential of artificial intelligence in nuclear waste management: Applications, challenges, and future directions	2025	Christopher Selvam, Damian; Devarajan, Yuvarajan; Raja T
8	Artificial intelligence application in healthcare management: a review of challenges	2024	Sorkhi, Abbas ; Pourasghar, Faramarz
9	Recent advances in applications of artificial intelligence in solid waste management: A review	2022	Ihsanullah I; Alam, Gulzar; Jamal, Arshad; Shaik, Feroz
10	Urgent need for developing a framework for the governance of AI in healthcare	2020	Baig, Mansoor Ali; Almuhaizea, Mohamad A; Alshehri, Jumanah; Bazarbashi, Mohammad Shouki; Al-Shagathrh, Fahad
11	Smart waste management: a systematic review and scientometric analysis of artificial intelligence applications	2025	Singh, Jaiteg; El-Sappagh, Shaker; Ali, Farman; Goyal S.B; Kumar, Manoj
12	AI's role in the implementation of circular	2025	Khulbe, Manjula

	economy practices: Optimising product life cycles, promoting recycling and facilitating the reuse of materials		
--	--	--	--

Applications of AI in Healthcare Waste Management

The integration of Artificial Intelligence (AI) in healthcare waste management is emerging as a promising solution to address the growing challenges of medical waste (MW) management. Many aspects will be discussed in the implementation of AI in healthcare waste management, such as identification, classification and tracking, smart waste management systems, integration with other technologies, and predictive analytics and monitoring.

Artificial intelligence (AI) technology plays a pivotal role in the modernization of medical waste management by facilitating real-time monitoring and predictive analytics. Through the integration of machine learning algorithms and computer vision, AI systems can accurately classify various categories of medical waste, including infectious, hazardous, and recyclable materials [8]. This precise classification reduces human error and ensures compliance with regulatory standards, thereby mitigating the risk of contamination and enhancing the overall safety. Furthermore, AI-powered tracking systems enable monitoring of waste movement from the point of generation to final disposal, ensuring transparency and accountability throughout the waste management lifecycle.

In addition, AI optimizes the logistical operations associated with medical waste collection and treatment. By analyzing data on waste generation patterns, AI can predict peak periods and recommend efficient routing for waste collection vehicles, thereby reducing fuel consumption and operational costs. Within treatment facilities, AI supports the automation of sorting and processing tasks, thereby enhancing throughput and minimizing manual labor. These advancements not only contribute to environmental sustainability by reducing carbon emissions and promoting recycling but also assist healthcare institutions in more effectively complying with stringent waste management regulations.

Artificial intelligence (AI) has the potential to significantly enhance the efficiency of waste management systems by optimizing waste collection, sorting, and disposal methods. For instance, AI models can reduce transportation distances by up to 36.8%, achieve cost savings of up to 13.35%, and realize time savings of up to 28.22% [9]. These advancements are facilitated through the integration of machine learning algorithms and real-time data analytics, which enables dynamic route planning based on traffic patterns, waste generation rates, and bin fill levels. By continuously analyzing data from sensors and GPS-enabled vehicles, AI systems can prioritize collection points, avoid unnecessary trips, and ensure timely waste pickup, thereby enhancing operational efficiency and reducing the environmental impact [9]. Furthermore, AI-driven automation in waste sorting facilities can significantly increase the accuracy and speed of material separation. Advanced image recognition and robotic systems can identify and sort recyclables from mixed waste streams with greater precision than manual methods, thereby reducing contamination and improving the quality of recovered materials. This not only supports more sustainable recycling

practices, but also lowers labor costs and enhances worker safety by minimizing human exposure to hazardous materials [8].

The integration of artificial intelligence (AI) with the Internet of Things (IoT) and blockchain technologies has the potential to transform healthcare waste management by establishing a comprehensive and interconnected system. IoT devices can be strategically deployed throughout the waste management chain, from collection points to disposal facilities, to provide real-time data on waste volume, type, and location [10]. These data can be processed using AI algorithms to analyze patterns, predict waste generation trends, and optimize collection routes and schedules [11]. Furthermore, the AI system can monitor the performance of waste treatment equipment, anticipate maintenance requirements, and recommend proactive measures to prevent equipment failure. The blockchain technology can enhance this system by adding a layer of security and transparency. By recording each step of the waste management process on an immutable ledger, the blockchain ensures that all stakeholders have access to accurate and tamper-proof information. This feature is particularly valuable for tracking hazardous waste, ensuring regulatory compliance, and verifying appropriate disposal methods [10]. The integration of AI, IoT, and blockchain can thus create a more efficient, accountable, and sustainable healthcare waste management system, ultimately reducing the environmental impact and improving public health outcomes [10,12].

AI-driven predictive analytics has the potential to substantially enhance waste management systems by utilizing real-time data and historical trends to forecast waste generation patterns, identify potential risks, and optimize collection schedules. In the realm of hazardous and toxic waste, these predictive models can anticipate increases in waste production, detect anomalies that may suggest illegal dumping or system failures, and trigger alerts for immediate responses. This proactive approach not only mitigates environmental and health hazards but also enhances regulatory compliance and resource allocation by enabling authorities to act preemptively before issues escalate [13].

Moreover, the integration of AI with sensor networks and geospatial data facilitates precise tracking of hazardous waste from its origin to its final disposal. This comprehensive visibility ensures that waste is managed by following safety protocols, thereby reducing the likelihood of contamination or accidental exposure. Predictive analytics can also aid strategic planning by identifying high-risk zones, optimizing waste transport routing, and informing infrastructure investments. Overall, the application of AI in this domain improves operational efficiency, supports sustainable practices, and bolsters public health safeguards [13,14].

Challenges and Considerations

Although there were many benefits related to the implementation of AI in healthcare waste management, there were many challenges and considerations, such as data quality and accessibility, ethical and social concerns, and regulatory and governance frameworks. A primary challenge in implementing AI in healthcare waste management is ensuring the availability and quality of data. Accurate and comprehensive data are crucial for developing reliable AI algorithms. This challenge stems from several factors, including the diverse nature of healthcare waste, inconsistent reporting practices across healthcare facilities, and the complexity of waste management processes [15]. Healthcare waste encompasses a wide range of materials, from general waste to hazardous and infectious substances, each

requiring specific handling and disposal methods. The heterogeneity of this waste makes it difficult to standardize the data collection and analysis processes[15].

Furthermore, the implementation of AI in healthcare waste management requires not only historical data but also real-time information to make accurate predictions and optimize waste-handling procedures. This necessitates the integration of various data sources such as waste generation rates, disposal costs, and environmental impact assessments. Ensuring the interoperability of different data systems and maintaining data privacy and security add additional layers of complexity to the data challenge. Overcoming these obstacles requires collaboration between healthcare providers, waste management companies, and technology developers to establish standardized data collection protocols and create robust data infrastructure that can support AI-driven waste management solutions [15].

The integration of artificial intelligence (AI) in healthcare waste management offers substantial potential for enhancing efficiency and safety. However, this necessitates a careful examination of ethical and social considerations. A primary concern is data privacy, as AI systems frequently depend on extensive datasets that may contain sensitive patient or institutional information [11]. Adherence to data protection regulations and implementation of robust cybersecurity measures are crucial for maintaining trust and safeguarding personal information. Furthermore, algorithmic bias poses a risk of inequitable outcomes if AI models are trained on non-representative or biased data, potentially disadvantaging certain populations or healthcare facilities. To mitigate these risks and promote fairness, transparent model development, regular audits, and inclusive data practices are essential [15].

Additionally, the environmental impact of AI technologies employed in healthcare waste management warrants further attention. The energy consumption associated with training and operating AI models, particularly those that require substantial computational power, contributes to carbon emissions and resource depletion. This environmental cost may undermine the sustainability objectives that the AI seeks to support in waste management. Consequently, the adoption of energy-efficient algorithms, utilization of green computing practices, and evaluation of the life cycle impact of AI systems are necessary to ensure that technological advancements align with broader environmental and social responsibility goals [11], [15].

Establishing robust governance and compliance frameworks is crucial for ensuring the efficacy, safety, and ethical considerations of AI applications in healthcare waste management [12]. These frameworks form the foundation for setting clear standards, protocols, and accountability mechanisms that guide the development, deployment, and monitoring of AI technology. By enforcing regulatory compliance and ethical guidelines, stakeholders can mitigate risks, such as data breaches, algorithmic bias, and unintended environmental consequences. Furthermore, governance structures help to ensure that AI systems are transparent, explainable, and aligned with public health objectives and environmental sustainability goals.

This collaborative approach not only enhances the reliability and safety of AI applications, but also ensures that their implementation respects patient privacy, adheres to environmental regulations, and supports equitable access to technological benefits. Ultimately, such frameworks are critical for fostering responsible innovation and long-term success in integrating AI into healthcare waste management practices [12].

Future Directions

Ongoing research is essential to overcome the limitations and challenges associated with AI in healthcare waste management. To fully harness the transformative potential of AI in this domain, it is crucial to address multifaceted challenges impeding its widespread adoption. Enhancing data quality is fundamental because AI systems depend significantly on accurate, comprehensive, and timely data to produce reliable insights. This necessitates the standardization of data collection methods, ensuring system interoperability and safeguarding data privacy and security [15].

Policy and multi-sector collaboration are critical in facilitating AI integration. Policymakers should establish clear regulatory frameworks that support innovation, while addressing concerns related to accountability, data governance, and environmental impact. Incentivizing research and development, providing funding for pilot projects, and creating platforms for knowledge exchange can expedite progress in this field [12]. Collaboration among technologists, healthcare providers, waste management companies, and consumers is vital for co-creating solutions that are both technologically robust and socially acceptable [12]. By fostering a collaborative ecosystem and addressing existing barriers, stakeholders can unlock the full potential of AI in transforming healthcare waste management into a more efficient, cost-effective, and environmentally responsible process.

4 Conclusion

Artificial intelligence possesses considerable potential to revolutionize healthcare waste management by improving efficiency, reducing costs, and promoting sustainability. Nonetheless, it is crucial to address the challenges related to data quality, ethical considerations, and regulatory frameworks to ensure the successful integration of AI in this sector. Ongoing research and collaboration among stakeholders will be essential to fully harness the benefits of AI in managing healthcare waste.

References

- [1] Janik-Karpinska E, Brancaloni R, Niemcewicz M, Wojtas W, Foco M, Podogrocki M, et al. Healthcare Waste-A Serious Problem for Global Health. *Healthcare (Basel, Switzerland)*. 2023;11(2): 242. <https://doi.org/10.3390/healthcare11020242>.
- [2] Thakur V, Mangla SK, Tiwari B. Managing healthcare waste for sustainable environmental development: A hybrid decision approach. *Business Strategy and the Environment*. 2021 Jan;30(1):357-73.
- [3] Zhang HJ, Zhang YH, Wang Y, Yang YH, Zhang J, Wang YL, Wang JL. Investigation of medical waste management in Gansu Province, China. *Waste Management & Research*. 2013 Jun;31(6):655-9.
- [4] Padmanabhan, K.K.; Barik, D. Health Hazards of Medical Waste and Its Disposal. In *Energy from Toxic Organic Waste for Heat and Power Generation*; Elsevier: Amsterdam, The Netherlands, 2019; pp. 99–118
- [5] Yazie, T.D.; Tebeje, M.G.; Chufa, K.A. Healthcare Waste Management Current Status and Potential Challenges in Ethiopia: A Systematic Review. *BMC Res. Notes* 2019, 12, 285

- [6] Soni U, Roy A, Verma A, Jain V. Forecasting municipal solid waste generation using artificial intelligence models—a case study in India. *SN Applied Sciences*. 2019 Feb;1(2):162.
- [7] Wang K, Zhao Y, Gangadhari RK, Li Z. Analyzing the adoption challenges of the Internet of things (Iot) and artificial intelligence (ai) for smart cities in china. *Sustainability*. 2021 Oct 3;13(19):10983.
- [8] Carrasco, Diego Cajamarca,Tobar-Ruiz, María Gabriela,López, Diego Marcelo Almeida,Hermida, Carlos Eduardo Cevallos,Arellano, Verónica Magdalena Llangari,Tobar, Mateo Augusto Zavala;Godoy, María Magdalena Paredes.Bibliometric analysis of the applicability of artificial intelligence in the integrated management of medical waste. *Hybrid Gold Open Access*.Volume 31 January 2024.
- [9] Fang B, Yu J, Chen Z, Osman AI, Farghali M, Ihara I, Hamza EH, Rooney DW, Yap PS. Artificial intelligence for waste management in smart cities: a review. *Environmental Chemistry Letters*. 2023 Aug;21(4):1959-89.
- [10]Bamakan SM, Malekinejad P, Ziaecian M. Towards blockchain-based hospital waste management systems; applications and future trends. *Journal of Cleaner Production*. 2022 May 15;349:131440.
- [11]Dewasiri NJ, Rathnasiri MS, Karunarathna KS. Artificial Intelligence-Driven Technologies for Environmental Sustainability in the Healthcare Industry. In*Transforming Healthcare Sector Through Artificial Intelligence and Environmental Sustainability* 2025 Jan 23 (pp. 67-87). Singapore: Springer Nature Singapore.
- [12]Sharma V, Jamwal A, Agrawal R, Pratap S. A review on digital transformation in healthcare waste management: Applications, research trends and implications. *Waste Management & Research*. 2025 Jun;43(6):828-49.
- [13]Chenniappan T, Devarajan Y. A critical analysis of the role of artificial intelligence and machine learning in enhancing nuclear waste management. *Kerntechnik*. 2024 Oct 28;89(5):586-94.
- [14]Selvam DC, Devarajan Y, Raja T. Exploring the potential of artificial intelligence in nuclear waste management: Applications, challenges, and future directions. *Nuclear Engineering and Design*. 2025 Jan 1;431:113719.
- [15]Sorkhi A, Pourasghar F. Artificial intelligence application in healthcare management: a review of challenges. *International Journal of Healthcare Technology and Management*. 2024;21(1):79-88.